Analysis of Learning Assessment Role Using Active Methodologies in "KAA" Perspective.

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Abstract

This article has in context the proposition of a discipline of Problem-Based Learning (PBL) in which it seeks to achieve soft skills competencies, while technical content is transmitted in the area of product engineering. Through a bibliometric research This analysis seeks to map the studies that encompass the main concepts around the theme: performance evaluation with the use of active methodologies for students from various engineering courses, in particular: computer, software, electrical and mechatronics. The publications analyzed were located through consultations with the database of Scopus journals, being considered as cutting papers published between the years 2007 to 2018. In the classification of the studies, special attention was paid to the performance evaluation process with a focus on the perspective do knowledge, ability and attitudes (KAA). After the definition of the sample, content analysis was performed in 20 articles focusing on the mapping of the evaluation processes of performance in order to identify the most recurrent practices in the evaluative process. The results indicate that the analyzed studies followed the same tendency in their evaluation systems by means of the evaluation method with questionnaires that measure a priori the self-perception of the student when subjected to the active learning dynamics, combined with Application of examinations or tests in order to assess the degree of understanding by means of grades.

Keywords: Active Learning; Evaluating, Problem-Based Learning; Soft Skills.

1 Introduction

The evaluative practices in active education methodologies, especially in engineering, are still elements to be better consolidated. On the one hand, the perception assessments of students and educators are aligned with the conception of education used in these practices. On the other hand, the technical content and the demand for professionals already qualified for immediate use in the market points to more pragmatic tests about the content of technical engineering disciplines.

However, before we deal with the evaluation question, it is paramount that the pedagogical approaches based on expository classes remain dominant in engineering education. This leaves graduates poorly prepared to understand the complexities of the profession (ASEE, 2003). Specifically, the approach based on expository classes emphasizes procedural knowledge and does not have the ability to keep the students' attention, which leads to low frequency rates (Mills & Treagust, 2003). This means that students may not feel motivated to go to classes nor will they be retaining information from classrooms that emphasize memorization and recollection. This is problematic given that the emphasis on memorization rather than application of the knowledge does not provide students with opportunities to see the real use of their learning. Which, in turn, contributes to the high evasion rate verified in the engineering courses? In Brazil, rates reach almost 50% according to the Brazilian Association of Engineering Education (Abenge, 2018).

A report by the American Society for Engineering Education (ASEE, 2009) highlights that to address the growing list of interconnected and complex global problems, it is necessary to train engineers that are capable of dealing with the multifaceted nature of the challenges to be faced in the 21st century. In a recent motion for a resolution on the curricular guidelines for Brazilian engineering courses elaborated by the (i) Enterprise mobilization for Innovation (MEI), (ii) National Confederation of Industry (CNI) and (iii) Abenge entitled "Innovation in Engineering Education: a proposal of national curriculum guidelines for the engineering courses ", it is noted that among the main aspects contained in this document the suggestion that the training in

engineering courses changes from training based on content to training based on skills. As this has occurred in courses that have become international reference in other countries. This means that the most important is not just knowing (content) but knowing how to do the task considering attitudes and ethical behaviors. In order to happen, the exchange of traditional classrooms for learning environments, in which contextualized activities are developed based mainly on active learning (Abenge, 2018).

It is believed that the traditional approach of classes does not demand high social skills from students, such as communication, collaboration, personal skills and design skills (Nguyen, 1998). This article does not aim to extend the multiple causes of success or failure in the professional career of engineers. The careers are no longer as predictable as they were in the past, and it is quite likely that an egress today will act in various sectors of the economy, in companies of different sizes and types (Escrivão Filho & Ribeiro, 2009). Thus, it is perceived that engineering students at the "world" level cannot be taught how to deal with the problems of tomorrow and therefore it is imperative to review and direct the undergraduate learning experience in engineering.

Turning to the question of the motivational object performance evaluation of this research, we started with the approach proposed by Bloom (1993), in which evaluation can be characterized in three ways: (i) Diagnostic Evaluation (analytical), (ii) the formative Assessment (parent) and (iii) the summative Evaluation (classificatory). It is possible to observe that the diagnostic evaluation has three objectives: the first is to identify the reality of each student who will participate in the process; the second is to check whether the student presents ability and pre-requisites for the process; the third objective is related to the identification of causes, of recurrent difficulties in learning. Thus, it is possible to review the educational action to remedy the problems.

The formative evaluation is the one that has the function to control and, as such, should be performed throughout the school period. Aiming to verify whether the students are achieving the objectives previously proposed. This function of the evaluation searches, basically, to evaluate whether the student dominates gradually and hierarchically each stage of learning, before advancing to another subsequent stage of teaching-learning. We can also say that it is motivating because it avoids the tensions caused by traditional assessments. It is with the formative evaluation that the student gains knowledge from their errors and successes. Also, the student is encouraged to continue their studies systematically. The formative evaluation allows the professor to detect and identify gaps in the way of teaching, assisting in the reformulation of his didactic work, aiming to improve it. The summative evaluation has as basic function the classification of the students. It is performed at the end of a course or teaching unit to classify students according to their previously established levels of use. Currently, the classification of students is processed according to their income, based on the objectives foreseen. According to Bloom (1983), the summative evaluation "aims to evaluate in a general way the degree to which the broader results have been achieved along and at the end of a course". Through this type of assessment, the students' feedback that is provided informs how much learning was achieved. This is the central objective of the formative evaluation and it lends itself to the comparison of results obtained, also aiming at the attribution of grades. These three functions of the evaluation should be linked or adjusted to ensure the efficiency and effectiveness of the evaluation system and thus having as a result the excellence of the teaching-learning process. Although there is no unanimous agreement on the meaning of "competence," many authors define this as the ability to perform specified tasks effectively (Veraldo, 2017).

To Leme (2012), the competence consists of the following elements: knowledge, ability and attitude, thus defined: (i) knowledge is knowing, (ii) ability is the know-how and (iii) attitude is what leads us to exercise our ability of a given knowledge, because it is the desire to do, an abstract characteristic not observable. Rabaglio (2001), explains that the meanings for the acronym KAA, can be described as: knowledge (K): It concerns knowledge acquired in the course of life, in schools, universities, courses; ability (A): consists of the ability to perform a particular task, physical or mental; attitude (A): Related to behaviors in face of everyday situations and tasks.

According to Reis, Barbalho & Zanetti (2017), engineering education has been the object of studies in search of approaches that provide better results in terms of learning. There are countless publications on the use and analysis of the PBL in consultations in the most diverse databases. The results in general highlight the benefits

of using the approach to learning by providing greater absorption of technical content by students and the development of multidisciplinary abilities.

However, this analysis seeks to map a survey of considerations around the theme: performance evaluation with the use of PBL for students coming from various engineering courses. Publications of interest were located through consultations to periodic database "Scopus" and "Web of Science" through the journal for the improvement of higher education personnel being considered as clipping the works published between the years of 2007 and 2018. In the classification of the work, special attention was directed to the performance evaluation process with a focus on the KAA perspective. From this perspective, we want to propose a meeting among the most commonly cited evaluation instruments among the articles analyzed taking into consideration the active methodology PBL as well as the three dimensions of competence based on Durand (2000).

In addition to this introduction, the work has other three parts. Topic 2 presents the extract from the Bibliometric analysis of some researches with the application of the PBL in teaching Engineering. Topic 3 discusses issues related to the evaluation of the four skills in the perspective of the three dimensions of the KAA applied to PBL. Topic 4 brings final considerations to results that came after the development of the research.

2 Bibliometric Analysis

To select the articles, a methodological approach of bibliometric research was adapted to the interests of the research. First, the following filters were directed to the Scopus collection (Elsevier) with the following criteria:

- Topics: Project method in teaching, project-based learning, educational technology, active learning, teaching methods, and problem-based learning resulting in 3,417 articles from 2007 to 2018 when filtered only the following keywords: PBL and Active learning.
- Adjusting the filter to: PBL, active learning OR learning by doing; articles in the Journal Title IEEE transactions on education; peer-reviewed periodicals; centered from 2007 to 2018; cut out sorted by relevance.

From this extraction the 12 articles of the ranking are detailed in table 1.

Table 9. Detailing articles the IEEE Transaction on Education Journal

Title	Authors	Year
A Blended Learning Approach to Course Design and Implementation	Hoic-Bozic, Mornar, & Boticki, 2009.	
Active Methodologies in a Queueing Systems Course for Telecommunication Engineering Studies	García & Hernández, 2010	
Engaging High School and Engineering Students: A Multifaceted Outreach Program Based on a Mechatronics Platform	Habash & Suurtamm, 2010.	
Effective Teaching of the Physical Design of Integrated Circuits Using Educational Tools	Mahfuzul, Sicard, & Dhia, 2010.	
Project-Based Learning and Rubrics in the Teaching of Power Supplies and Photovoltaic Electricity	Martinez, Herrero, & de Pablo, 2011.	
Experiences in the Application of Project-Based Learning in a Switching-Mode Power Supplies Course	Lamar et al, 2012.	
Application of Project-Based Learning (PBL) to the Teaching of Electrical Power Systems Engineering	Hosseinzadeh & Hesamzadeh, 2012.	
A Smart Home Test Bed for Undergraduate Education to Bridge the Curriculum Gap from Traditional Power Systems to Modernized Smart Grids	Hu, Li, & Chen, 2015.	
Analysis of Introducing Active Learning Methodologies in a Basic Computer Architecture Course	Arbelaitz, Martin, & Muguerza, 2015.	
Using PBL to Improve Educational Outcomes and Student Satisfaction in the Teaching of DC/DC and DC/AC Converter	Martinez-Rodrigo et al., 2017.	
A Multidisciplinary PBL Approach for Teaching Industrial Informatics and Robotics in Engineering	Calvo et al., 2018.	
Hybrid Problem-Based Learning in Digital Image Processing: A Case Study	Tan & Shen, 2018.	

The filter was then directed to the collection "Web of Science" with the consecutive caveats:

- Problem-Based Learning (PBL), Active learning, engineering and applied, articles from journals, peer-reviewed journals, concentered between 2007 and 2018; cut out performed by number of citations of each article and relevance.

This search resulted in 49 articles of which the 8 first articles of the ranking detailed in table 2.

Table 2. Articles analyzed from several journals.

Title	Authors	Source
Enabling and KAAracterizing Twenty-First Century Skills in New Product Development Teams	Cobb et al., 2008.	International Journal of Engineering Education
Problem-based learning for design and engineering activities in virtual worlds	Vosinakis & Koutsabasis, 2012.	PRESENCE: Teleoperators And Virtual Environments
Experience applying language processing techniques to develop educational software that allow active learning methodologies by advising students	Castro-Schez et al., 2014.	Elsevier - Journal of Network and Computer Applications
Engineering Students: Enhancing Employability Skills through PBL	Othman, et al., 2017.	IOP Conference Series: Materials Science and Engineering. IOP Publishing,
Problem-based learning of process systems engineering and process integration concepts with metacognitive strategies: The case of P-graphs for polygeneration systems	Promentilla et al.,2017.	Elsevier - Applied Thermal Engineering
Incorporation of Sustainability Concepts into a Civil Engineering Curriculum	Chau, 2007.	Journal of Professional Issues in Engineering Education and Practice
Applying Active Methodologies for Teaching Software Engineering in Computer Engineering	Fonseca & Gómez, 2017.	IEEE Revista Iberoamericana De Tecnologias del Aprendizaje
Incorporating life cycle assessment and ecodesign tools for green chemical engineering: A case study of competences and learning outcomes assessment	Margallo, Dominguez-Ramos, & Aldaco, 2018.	Education for Chemical Engineers

In this context, we conducted a study of the results of each article and its respective performance evaluation approaches after applying the PBL methodology during the teaching learning process in undergraduate and post-graduation courses-graduation level of masters both in engineering, considering the following categories: (i) competences according to Durand (2000): knowledge, ability and attitude (ii) evaluation elements/instruments during the learning process (projects, proof, seminar, questionnaire, etc.) and added as reflection the (iii) classification of the instrument according to Bloom (1993): diagnostic, formative or summative.

3 Competences

The perspective of the KAA or the competency assessment is being used in this work as a dividing line in the evaluative context when proposed by the PBL methodology believing that the active methodologies cannot be applied in the educational scope separately for a regular higher education system. Based on this proposal, the evaluative process of the articles analyzed by this study were classified in Table 3, which includes the evaluation of knowledge, ability and attitude, the elements evaluated and the most commonly evaluation instruments used in the set of articles are separated by the focus given in the type of competence. We identified 10 articles where it is possible to establish that 3 dimensions of the KAA were an agent of concern during the implementation of the performance evaluation (Hoic-Bozic, Mornar, & Boticki, 2009; Martinez, Herrero, & de Pablo, 2011; Lamar et al, 2012; Haje & Hesamzadeh, 2012; Hu, Li, & Chen, 2015; Arbelaitz, Martin, & Muguerza, 2015; Martinez-Rodrigo et al., 2017; Calvo et al., 2018; Cobb et al., 2008; Vosinakis & Koutsabasis, 2012) with a more objective focus on developing the threefold skills, although not all of them leave the three dimensions aligned with their evaluative instruments.

Another 6 articles had a more evaluative approach focused only on the Knowledge (García & Hernández, 2010; Habash & Suurtamm, 2010; Mahfuzul, Sicard, & Dhia, 2010; Othman, et al., 2017; Castro-Schez et al., 2014; Fonseca & Gómez, 2017) although some Ability and Attitudes are cited in their learning objectives. In the same line of approach, two other articles (Promentilla et al.,2017; Chau, 2007) indicate greater targeting in Abilities and Attitudes, that is, when we measured their evaluative instruments, we hoped to find formative and diagnostic assessments rather than simply summative, where written exams are predominant. And last Tan & Shen (2018), they present in their work an approach focused on Knowledge and Ability, not explaining Attitudes in its evaluation. Margallo, Dominguez-Ramos, & Aldaco, 2018 certainly presented a methodology for performance evaluation very close to the strategy we hope to adopt in the context of an active methodology concerned with the 3 aspects of the competence of the KAA.

3.1 Knowledge

According to (Zamyatina et al, 2014), engineering graduates should demonstrate "Knowledge" basic principles related to the engineering of his profession; specialization in development and use of new products and systems, as well as understanding the importance and strategic value of the technological development of society. As already quoted, according to Leme (2012), the "Knowledge" is the know-how. It refers to the study in schools, books, work, and life. Rabaglio (2001), describes the "Knowledge", as the information that accumulates throughout life, through the study. However, in order to obtain the "Knowledge" coming from the study in higher education, the courses generally organize themselves in various disciplines, which together at the end must subsidize the training of professionals for the labor market (technical training). In the articles evaluated, it was observed that the perspective of "Knowledge" had as convergence among the analyzed articles the choice of the most traditional evaluation, for example theoretical exams.

3.2 Abilities

According to Bloom et al (1983), "Ability" is associated with the capability of students to convert and use their knowledge about the discipline. Their knowledge of the world and its life experiences, as well as the techniques used to solve a current problem are interrelated, which makes the competence a set of skills that unites knowledge, abilities and attitudes over a certain area. In view of the concept of "Ability" as know-how (Rudder, 2012), it is understood as technical capacity and knowledge as (Durand, 2000). Teamwork was the ability most cited in the articles evaluated, followed by the ability of oral communication. However, several other abilities could be evaluated within the KAA perspective along with active methodologies as long as the educator manages to make use of other instruments such as a "logbook" containing photos in students ' miniatures so that they can take notes during and after the encounters of a wider range of abilities. Among other abilities, also cited less frequently by the authors, are: written communication, systemic vision, critical analysis, problem solving, independent study and self-regulated work.

3.3 Attitudes

Leme (2012), refers to attitude as being what leads us to exercise our ability of a given knowledge, because it is the desire to do. It is an abstract characteristic, not observable. This means it is related to behaviors in the face of situations and daily tasks, according to Rabaglio (2001). Among the articles evaluated, leadership was the attitude more quoted as an evaluative point, although this dimension did not have a broad coverage for the articles evaluated. However, other terms may have been mentioned. Table 3 details ten articles of the twenty analyzed for this study.

Table 3. Description and classification Assessment instruments and competences

Title	Authors	Competencies: KAA mentioned on paper		Assessment Tool
Incorporating life cycle assessment and ecodesign tools for green chemical engineering: A case study of competences	Margallo, Dominguez-Ramos, & Aldaco, 2018.	Knowledge	Eco-design Environmental Sustainability and Eco-design Relevance of Environmental Issues	1. Portfolio 2. Poster exhibition 3. Competency Acquisition Assessment Questionnaire
		Abilitys	Teamwork, Creativity; And Entrepreneurship	

and learning outcomes assessment		Attitude	Initiative	
Instrument classification: Formative – no examination was foreseen.				
Applying Active Methodologies for Teaching Software Engineering in Computer Engineering	Fonseca & Gómez, 2017.	Knowledge	Software Engineering	1. Software Design 2. Report 3. Presentation
		Abilitys	Team-Work Ability to Synthesize	
		Attitude	Autonomous Work Responsibility	
Instrument classification: Formative – no examination was foreseen.				
Incorporation of Sustainability Concepts into a Civil Engineering Curriculum	Chau, 2007.	Knowledge	Design Project to Civil Engineering	1. Student feedback questionnaire, 2. Peer review, 3. Supervisors' comments 4. Employer surveys.
		Abilitys	Resolução de problemas, Trabalho em equipe multidisciplinar, Comunicação escrita e verbal, Abilitys interpessoais, Gerenciamento de projetos	
		Attitude	Liderança	
Instrument Classification: Formative				
Experience applying language processing techniques to develop educational software that allow active learning methodologies by advising students	Castro-Schez et al., 2014	Knowledge	Language Processor techniques for designing and developing educational software tools	1. Exercises 2. Tool Satisfaction´s Questionnaire
		Abilities	Interpretação Raciocínio	
		Attitude	Not defined	
Instrument Classificatoin: Not specified				
Engineering Students: Enhancing Employability Skills through PBL	Othman, et al., 2017	Knowledge	Not defined	1. PBL Group versus Control Group
		Abilities	Fundamental Skills, Self-Management Skills, Team Work Skills.	
		Attitude	Responsibility Self-motivated Strong leadership qualities Strategic thinking abilities	
Instrument Classification: Not Applicable				
Active Methodologies in a Queueing Systems Course for Telecommunication Engineering Studies	García & Hernández, 2010	Knowledge	Queuing systems	1. Intermediate Examination (TEST) 2. PBL Rating (optional) 3. Questionnaire 4. Final Exam 5. Exercise resolution (problems)
		Ability	Not defined	
		Attitude	Not defined	
Instrument classification: Summative				
Problem-based learning for design and engineering activities in virtual worlds	Vosinakis & Koutsabasis, 2012	Knowledge	Functional and/or Digital prototype	Project (Group) Ability
		Abilities	Working Group Self-directed learning Self-assessment	
		Attitude	Project evaluation Critical Thinking Responsabiblity	
Instrument Classification: summative and formative evaluation				
Enabling and characterizing Twenty-	Cobb et al., 2008		Development of new products; Social entrepreneurship and	1. Satisfaction survey with graduates for

First Century Skills in New Product Development Teams		Knowledge	Socially conscious design projects.	(student leaders who worked with teams having a diverse mix of skills).
		Abilities	Teamwork, Generation and creativity of concepts, Communication abilities, Business abilities, project management	
		Attitude	Ethics; Sense of professionalism, leadership, Dynamism; Agility Resilience and flexibility; effective meetings and scheduling; Setting goals and Work with the mission.	
Instrument Classification: No applicable				
Using PBL to Improve Educational Outcomes and Student Satisfaction in the Teaching of DC/DC and DC/AC Converter	Martinez-Rodrigo et al., 2017	Knowledge	Technical projects, Simulation techniques; Programmatic Content PBL Methodology	1. Report 2. Peer Review, 3. Simulations 4. Project 5. Presentations 6. Theoretical examination, 7. troubleshooting, 8. Student Satisfaction-PBL evaluation[2]
		Abilities	Teamwork Troubleshooting	
		Attitudes	Leadership Initiative Participation	
Instrument classification: Summative				
A Multidisciplinary PBL Approach for Teaching Industrial Informatics and Robotics in Engineering	Calvo et al., 2018	Knowledge	Robotics	1. Classroom activities, 2. Final written examination, 3. Oral presentation and videos. Project Tasks, 4. Student Satisfaction-PBL evaluation, 5. Subtasks with virtual deliveries,
		Abilities	Teamwork	
		Attitudes	Autonomous and Team Proactivity.	
Instrument classification: Summative				
Application of Project-Based Learning (PBL) to the Teaching of Electrical Power Systems Engineering	Hosseinzadeh & Hesamzadeh, 2012	Knowledge	Project Management	1. PBL Online Quiz 2. Portfolio 3. Presentation 4. Written test, 5. Lab Practice Test 6. Self-Assessment, 7. Peer review
		Abilitys	Time Management, Teamwork, Communication abilities	
		Attitudes	Not defined	
Instrument classification: Summative				

4 Conclusion

Soft skills are the competencies of the 21st century recognized by the market working as the differential of the professional future. Among the best known, some already cited in the articles studied, are self-knowledge, communication, proactivity, emotional management, creativity and interpersonal ability. The term KAA, was used in this work for covering both the vision of Hard skill, as in the K for "Knowledge" – Bringing the vision of technical know-how – and softs skills, as in the A for "Abilities" and "Attitude". Thus, enabling a complete and fair view to measure performance having an increasing collection of adjustments for the engineers' formation.

However, we believe that we have gaps not only in a matter of adequacy in how assessment is done, but also in the formation of those who evaluate. These educators are not generally prepared for the evaluation of professional's abilities, although they acknowledge that engineering students need more than the so called

[2] Aplicabilidade da abordagem – PBL, aceitação dos alunos, quando há uma ferramenta AVA avalia-se também a percepção de usabilidade do ambiente VW.

"difficult abilities" to be prepared for their professional future. Perrenoud (2000), worried about this debate on the competences of educators, developed some references on the subject. It proposes a series of specific competencies grouped into families of fundamental competences for educators in the pedagogical process, in order to better perform and develop their actions in the educational field.

The need for an evaluation of learning to be coherent with the epistemological assumptions that guide a curriculum that uses active methodology was determinant for the, timely and somewhat broader, interest in the use of evaluative practices training for the offer of the study product discipline of this article. This evaluation format provides regular and periodic evaluation, throughout the educational process, to obtain data on the progress achieved and thus effecting the timely correction of the distortions observed and fill in the gaps detected.

Among the many lessons learned, perhaps the most important was that the attention that the educator should have when proposing an alignment between the following: The evaluation instrument used as a tool to "measure" the degree of understanding of the student and the defined or expected learning objectives that students reach. That is, many articles are daring to direct their teaching activities to an approach under the application of active methodologies, often with virtual environments, but they forget to establish the exact form in how to track and measure success or failure within the development process.

Thus, we have seen many interesting proposals of engineering education with PBL application, as we seek an alignment in the objective of learning and its assessment instruments, we realized that in the vast majority there were distortions in the definition of evaluation of a purely active performance: where knowledge, abilities and attitudes were being worked out through the active methodologies (in this specific work, PBL) and that evaluation instruments were able to measure their degree of achievement in the teaching-learning process.

Another point worth highlighting in the list of lessons learned is that we have seen that active learning combines with many evaluative instruments. In this regard it drew attention to the expressive number of options used for assessments. As future work, we will use the main skills and attitudes identified as well as good evaluative practices for application in the planning of process and product innovation discipline.

5 References

ABENGE, (2018). Inovação na Educação em Engenharia: Proposta de Diretrizes Curriculares Nacionais para o Curso de Engenharia, Proposta ABENGE. Brasília: ABENGE.

ARBELAITZ, Martin, & Muguerza (2015). Analysis of introducing active learning methodologies in a basic computer architecture course. IEEE Transactions on Education, v. 58, n. 2, p. 110-116.

ASEE, (2009). Creating a culture for scholarly and systematic innovation in engineering education: Ensuring U.S. engineering has the right people with the right talent for a global society, American Society for Engineering Education, Washington DC.

ASEE, (2012). Innovation with Impact: Creating a Culture for scholarly and systematic innovation in engineering education, American Society for Engineering Education, Washington DC.

AZIZ, S. M.; SICARD, E., & BEN DHIA, S. (2010). Effective teaching of the physical design of integrated circuits using educational tools. IEEE Transactions on Education, v. 53, n. 4, p. 517-531.

BLOOM, B. S; ENGELHART, M. D.; FURST, E. J.; HILL, W. H.; & KRATHWOHL, D. R. (1983). Taxonomia de objetivos educacionais: domínio cognitivo. São Paulo: Pioneira,

CALVO, Isidro et al. (2018). A multidisciplinary pbl approach for teaching industrial informatics and robotics in engineering. IEEE Transactions on Education, v. 61, n. 1, p. 21-28,

CASTRO-SCHEZ, Jose Jesus et al. (2014). Experience applying language processing techniques to develop educational software that allow active learning methodologies by advising students. Journal of Network and Computer Applications, v. 41, p. 65-79.

CHAU, K. W. (2007). Incorporation of sustainability concepts into a civil engineering curriculum. Journal of professional issues in engineering education and practice, v. 133, n. 3, p. 188-191.

COBB, Corie L. et al. (2008). Enabling and Characterizing twenty-first century skills in new product development teams. International Journal of Engineering Education, v. 24, n. 2, p. 420.

DURAND, T. (1998). Forms of incompetence. In: INTERNATIONAL CONFERENCE ON COMPETENCE-BASED MANAGEMENT, 4., Oslo: Norwegian School of Management.

DURAND, T. (2000). L'alchimie de la compétence. Revue Française de Gestion, n. 127, p. 84-102.

ESCRIVÃO FILHO, E., & RIBEIRO, L. R. de C. (2009). Aprendendo com PBL – aprendizagem baseada em problemas: relato de uma experiência em cursos de engenharia da EESC-USP. Minerva, v. 6, n.1, p. 23-30,

FONSECA, V. M. F.; & GÓMEZ, J. (2017). Applying active methodologies for teaching software engineering in computer engineering. IEEE Revista Iberoamericana de Tecnologias del Aprendizaje, v. 12, n. 3, p. 147-155,

GARCÍA, J., & HERNANDEZ, A. (2010). Active methodologies in a queueing systems course for telecommunication engineering studies. IEEE Transactions on Education, v. 53, n. 3, p. 405-412,

HABASH, R. W., & SUURTAMM, C. (2010). Engaging high school and engineering students: A multifaceted outreach program based on a mechatronic platform. IEEE transactions on education, v. 53, n. 1, p. 136-143,

HOIC-BOZIC, N.; MORNAR, V., & BOTICKI, I. (2009). A blended learning approach to course design and implementation. IEEE transactions on education, v. 52, n. 1, p. 19-30.

HOSSEINZADEH, N., & HESAMZADEH, M. R. (2012). Application of project-based learning (PBL) to the teaching of electrical power systems engineering. IEEE Transactions on Education, v. 55, n. 4, p. 495-501,

HU, Q.; LI, F., & CHEN, C. (2015). A smart home test bed for undergraduate education to bridge the curriculum gap from traditional power systems to modernized smart grids. IEEE Transactions on Education, v. 58, n. 1, p. 32-38

LAMAR, D. G. et al. (2012). Experiences in the application of project-based learning in a switching-mode power supplies course. IEEE Transactions on education, v. 55, n. 1, p. 69-77.

LEME, R. (2012). Aplicação prática de gestão de pessoas por competências: mapeamento, treinamento, seleção, avaliação e mensuração de resultados de treinamento. Rio de Janeiro: Qualitymark Editora,

MARGALLO, M.; DOMINGUEZ-RAMOS, R., & ALDACO, A. (2018). Incorporating life cycle assessment and ecodesign tools for green chemical engineering: A case study of competences and learning outcomes assessment. Education for Chemical Engineers,

MARTÍNEZ, F.; HERRERO, L. C., & DE PABLO, S. (2011). Project-based learning and rubrics in the teaching of power supplies and photovoltaic electricity. IEEE Transactions on Education, v. 54, n. 1, p. 87-96,

MARTINEZ-RODRIGO, F. et al. (2017). Using PBL to improve educational outcomes and student satisfaction in the teaching of DC/DC and DC/AC converters. IEEE Transactions on Education, v. 60, n. 3, p. 229-237,

MILLS, J., & Treagust, D. (2003). Engineering Education Is Problem-Based or Project-Based Learning the Answer. Australasian Journal of Engineering Education, 2003, pp. 2–16.

NGUYEN, D. Q. (1998). The essential skills and attributes of an engineer: A comparative study of academics, industry per-

OTHMAN, H. et al. (2017). Engineering Students: Enhancing Employability Skills through PBL. In: IOP Conference Series: Materials Science and Engineering. IOP Publishing.

PERRENOUD, P. (2000). Dez novas competências para ensinar. Artmed.

PROMENTILLA, M. A. B. et al. (2017). Problem-Based learning of process systems engineering and process integration concepts with metacognitive strategies: The case of P-graphs for polygeneration systems. Applied Thermal Engineering, v. 127, p. 1317-1325.

RABAGLIO, M. O. (2001). Seleção por Competências. 2ª edição – Editora: Educator, São Paulo,

REIS, A. C. B.; BARBALHO, S. C. M; ZANETTE, A. C.D. (2017). A bibliometric and classification study of Project-based Learning in Engineering Education. Production, v. 27.

TAN, S., & SHEN, Z. (2018). Hybrid Problem-Based Learning in Digital Image Processing: A Case Study. IEEE Transactions on Education, v. 61, n. 2, p. 127-135,

VERALDO J. L. G., (2017). Avaliação completa das competências do engenheiro no projeto interdisciplinar do curso de Engenharia de Produção. UNESP, Guaratinguetá.

VOSINAKIS, S., & KOUTSABASIS, P. (2012). Problem-Based learning for design and engineering activities in virtual worlds. PRESENCE: Teleoperators and Virtual Environments, v. 21, n. 3, p. 338-358,

ZAMYATINA, O. M., et al., (2014). "Information technologies in engineering education: project activity and competence assessment." SGEM2014 Conference on Psychology and Psychiatry, Sociology and Healthcare, Education. Vol. 3. No. SGEM2014 Conference Proceedings, ISBN 978-619-7105-24-7/ISSN 2367-5659, September 1-9, 2014, Vol. 3, 411-418 pp. STEF92 Technology